The conjunctiva is a thin, protective mucous membrane found lining the outer surface of the eyeball (Figure 10.13).

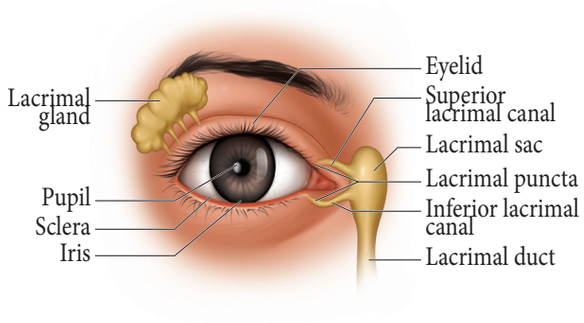


Figure.10.13 The human eye

The eye has two compartments, the **anterior** and **posterior compartments**. The anterior compartment has two chambers, first one lies between the cornea and iris and the second one lies between the iris and lens. These two chambers are filled with watery fluid called **aqueous humor**. The posterior compartment lies between the lens and retina and it is filled with a jelly like fluid called **vitreous humor** that helps to retain the spherical nature of the eye. **Eye lens** is transparent and biconvex, made up of long columnar epithelial cells called **lens fibres**. These cells are accumulated with the proteins called **crystalline**.

The eye ball

The eye ball is spherical in nature. The anterior one- sixth of the eyeball is exposed; the remaining region is fitted well into the orbit. The wall of the eye ball consists of three layers: fibrous **Sclera**, vascular **Choroid** and sensory **Retina** (Figure 10.14).

The outer coat is composed of dense non-vascular connective tissue. It has two regions: the anterior **cornea** and the posterior **sclera**. Cornea is a non-vascular transparent coat formed of stratified squamous epithelium which helps the cornea to renew continuously as it is very vulnerable to damage from dust. Sclera forms the white of the eye and protects the eyeball. Posteriorly the sclera is innervated by the optic nerve. At the junction of the sclera and the cornea, is a channel called **'canal of schlemm'** which continuously drains out the excess of aqueous humor.



Dilation and congestion of the blood vessels of the conjunctiva due to local irritation or infection are the cause of bloodshot eye (conjunctivitis - commonly called Madras eye). Infection of ciliary glands by bacteria causes a painful, pus filled swelling called a Sty.

The cornea is the only tissue in the body that can be transplanted from one person to another with little or no possibility of rejection. This is because cornea does not have blood vessels.

Choroid is highly vascularized pigmented layer that nourishes all the eye layers and its pigments absorb light to prevent internal reflection.
Anteriorly the choroid thickens to form the ciliary body and iris. Iris is the coloured portion of the eye lying between the cornea and lens. The aperture at the centre of the iris is the **pupil** through which the light enters the inner chamber. Iris is made of two types of muscles the **dilator papillae** (the radial muscle) and the **sphincter papillae** (the circular muscle).In the bright light, the circular muscle in the iris contract; so that the size of pupil decreases and less light enters the eye. In dim light, the radial muscle

TN_GOVT_XI_Bio Zoology_Ch10.indd 183 09-01-2020 5.31.39 PM